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Math 201 Final Exam Term 251

Original exam date: 16 December 2025. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. Let $f(x,y) = 2xy$ and R be the triangular region with vertices $(0,0)$, $(-3,-1)$, and $(0,-1)$. Then the average value of f over R is equal to

(a) $\frac{3}{2}$

(b) $\frac{9}{4}$

(c) $\frac{27}{8}$

(d) $\frac{8}{3}$

(e) $\frac{2}{3}$

2.

$$\int_0^{3/\sqrt{2}} \int_y^{\sqrt{9-y^2}} \sqrt{x^2+y^2}\, dx\, dy =$$

(a) $\frac{13\pi}{9}$

(b) $\frac{8\pi}{5}$

(c) $\frac{15\pi}{4}$

(d) $\frac{9\pi}{4}$

(e) $\frac{7\pi}{2}$

3. The maximum value of $f(x,y) = 6xy$, where $x > 0$ and $y > 0$, subject to the constraint $\frac{x^2}{4} + \frac{y^2}{9} = 1$ is equal to

(a) 36

(b) 47

(c) 18

(d) 26

(e) 44

4.

$$\int_0^3 \int_0^{\sqrt{9-x^2}} \int_0^{\sqrt{9-x^2-y^2}} \sqrt{x^2+y^2+z^2}\, dz\, dy\, dx =$$

(a) $\int_0^{\pi/2} \int_0^{\pi} \int_0^3 \rho^{5/2} \sin\phi \, d\rho \, d\phi \, d\theta$

(b) $\int_0^{\pi/2} \int_0^{\pi/2} \int_0^3 \rho^{5/2} \sin\phi \, d\rho \, d\phi \, d\theta$

(c) $\int_0^{\pi} \int_0^{\pi} \int_0^3 \rho^{3} \sin\phi \, d\rho \, d\phi \, d\theta$

(d) $\int_0^{\pi} \int_0^{\pi/2} \int_0^3 \rho^{3} \sin\phi \, d\rho \, d\phi \, d\theta$

(e) $\int_0^{\pi/2} \int_0^{\pi/2} \int_0^3 \rho^{3} \sin\phi \, d\rho \, d\phi \, d\theta$

5. If (a, b, c), with $a < 0$, is the point on the surface $4x^2 + 2y^2 - z^2 = 100$ where the tangent plane is parallel to the yz-plane, then $a^2 + b^2 + c^2 =$

(a) 100

(b) 256

(c) 16

(d) 25

(e) 225

6. The graph of the function

$$f(x, y) = 4x - 4xy + 2y^2 - 4y + x^3 + 10$$

has

(a) one saddle point and one relative minimum

(b) one saddle point and one relative maximum

(c) one relative minimum and one relative maximum

(d) one saddle point and two relative maxima

(e) no saddle point

7. If $R = \left\{(x, y) : 0 \le x \le \dfrac{\pi}{4},\ 0 \le y \le 1\right\}$, then

$$\iint_R 4\sqrt{2}\, y \cos x \, dA =$$

(a) $4\sqrt{2}$

(b) 2

(c) $\dfrac{\sqrt{2}}{4}$

(d) 4

(e) $\dfrac{\sqrt{2}}{2}$

8. The area of the region that lies inside $r = 1 + \cos\theta$ and outside $r = \cos\theta$ is

(a) $\frac{\pi}{2}$

(b) $\frac{7\pi}{4}$

(c) $\frac{5\pi}{2}$

(d) $\frac{3\pi}{2}$

(e) $\frac{5\pi}{4}$

9. If D is the distance between the point $Q(0, -2, 2)$ and the line given by the set of parametric equations $x = t,\ y = 2t - 3,\ z = 2t$, then $D =$

(a) 1

(b) 2

(c) 8

(d) 9

(e) 3

10.

$$\int_0^4 \int_0^y f(x, y)\, dx\, dy =$$

(a) $\int_0^4 \int_4^x f(x, y)\, dy\, dx$

(b) $\int_{-4}^0 \int_x^4 f(x, y)\, dy\, dx$

(c) $\int_{-4}^0 \int_0^x f(x, y)\, dy\, dx$

(d) $\int_0^4 \int_{4x}^0 f(x, y)\, dy\, dx$

(e) $\int_0^4 \int_x^4 f(x, y)\, dy\, dx$

11. The maximum value of $f(x, y) = 4x^2 + 4y^2 - 8y$ over the region in the xy-plane bounded by the graphs of $y = 2x^2$ and $y = 2$ is equal to

(a) -2

(b) 0

(c) 28

(d) 2

(e) 4

12. If (a, b, c) is the point on the paraboloid $z = x^2 + y^2$ that is closest to the point $(3, 3, 1)$, then $a^3 + b^3 + c^3 =$

(a) $64 - 4\sqrt{2}$

(b) $64 + 4\sqrt{2}$

(c) 0

(d) 10

(e) 6

13. The highest point on the curve of intersection of the cone $x^2 + y^2 - z^2 = 0$ and the plane $x + 4z = 9$ is

(a) $(-4, 0, 4)$

(b) $(1, \sqrt{3}, 2)$

(c) $(5, 2, 1)$

(d) $(3, -3, 3)$

(e) $(-3, 0, 3)$

14. The volume of the solid region bounded by the graphs of the equations

$$z = 4 - x^2 - y^2 \quad \text{and} \quad z = 4 - 2x$$

is equal to

(a) $\int_0^2 \int_{-\sqrt{1-(x-1)^2}}^{\sqrt{1-(x-1)^2}} \left(2x - x^2 - y^2\right) dy\, dx$

(b) $\int_0^2 \int_{-\sqrt{1-(x+1)^2}}^{\sqrt{1-(x+1)^2}} \left(x^2 + y^2 + 2x\right) dy\, dx$

(c) $-\int_{-2}^0 \int_{-\sqrt{1-(x-1)^2}}^{\sqrt{1-(x-1)^2}} \left(x^2 + y^2 + 2x\right) dy\, dx$

(d) $\int_{-2}^2 \int_0^{\sqrt{1-(x+1)^2}} \left(2x - x^2 - y^2\right) dy\, dx$

(e) $\int_0^2 \int_{-\sqrt{1+(x-1)^2}}^{\sqrt{1+(x-1)^2}} \left(2x - x^2 - y^2\right) dy\, dx$

15. Let (a, b, c) be a point on the line

$$x = 3t + 1, \quad y = t - 1, \quad z = 4t + 2.$$

If the distance from this point to the plane $2x+2y+z = 2$ is 8, then a possible value of $a^2+b^2+c^2 =$

(a) 79

(b) 12

(c) 97

(d) 150

(e) 158

16. The point with rectangular coordinates $(x, y, z) = (\sqrt{3}, 1, 2\sqrt{3})$ is represented in spherical coordinates by $(\rho, \theta, \phi) =$

(a) $\left(4, \frac{\pi}{6}, \frac{\pi}{3}\right)$

(b) $\left(4, \frac{7\pi}{6}, \frac{\pi}{6}\right)$

(c) $\left(4, \frac{\pi}{6}, \frac{7\pi}{6}\right)$

(d) $\left(4, \frac{\pi}{6}, \frac{\pi}{6}\right)$

(e) $\left(4, \frac{\pi}{3}, \frac{\pi}{6}\right)$

17.

$$\int_0^{\sqrt{\pi/2}} \int_x^{\sqrt{\pi/2}} \int_2^{12} \sin\left(y^2\right)\, dz\, dy\, dx =$$

(a) $12 - \sqrt{\frac{\pi}{2}}$

(b) 10

(c) 5

(d) $2\sqrt{\pi} + 12$

(e) $2\sqrt{\frac{\pi}{2}} + 12$

18. Let L be the arc length of the curve represented by the parametric equations

$$x = ae^{-t}\cos t, \qquad y = ae^{-t}\sin t, \qquad 0 \le t \le \frac{\pi}{2}.$$

If $L = 4\left(1 - e^{-\frac{\pi}{2}}\right)$, then $a =$

(a) $\sqrt{18}$

(b) $\sqrt{8}$

(c) 4

(d) $\sqrt{2}$

(e) 2

19. If $w = x^2 + y^2 + z^2$, $x = t\sin s$, $y = t\cos s$, $z = st^2$, then the value of $\dfrac{\partial w}{\partial t}$, when $s = 3$ and $t = 2$, is equal to

(a) $216 + \dfrac{\pi^2}{4}$

(b) $220 + \dfrac{\pi^2}{9}$

(c) 220

(d) 292

(e) $288 + \dfrac{\pi^2}{8}$

20. The volume of the solid region bounded below by the upper nappe of the cone $z^2 = x^2 + y^2$ and above by the sphere $x^2 + y^2 + z^2 = 4$ is equal to

(a) $\dfrac{26\pi}{3}\left(2 - \sqrt{2}\right)$

(b) $9\pi\left(2 - \sqrt{2}\right)$

(c) $\dfrac{8\pi}{3}\left(5 + \sqrt{2}\right)$

(d) $\dfrac{8\pi}{3}\left(2 - \sqrt{2}\right)$

(e) $\dfrac{8\pi}{3}\left(2 + \sqrt{2}\right)$

21. The surface of a mountain is modeled by the equation

$$h(x, y) = 5000 - 0.001x^2 - 0.004y^2.$$

A mountain climber is at the point $(500, 300, 4390)$. Which of the following vectors points in the direction of steepest descent (descend at the greatest rate)?

(a) $5\mathbf{i} - 12\mathbf{j}$

(b) $5\mathbf{i} + 12\mathbf{j}$

(c) $-5\mathbf{i} + 3\mathbf{j}$

(d) $-5\mathbf{i} - 3\mathbf{j}$

(e) $-\mathbf{i} - 4\mathbf{j}$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	Integral $= \frac{9}{4}$, area $= \frac{3}{2}$, average $= \frac{3}{2}$.
2	**(d)**	Polar: $0 \le \theta \le \frac{\pi}{4}$, $0 \le r \le 3$: $\int r^2\, dr\, d\theta = \frac{9\pi}{4}$.
3	**(c)**	Lagrange: $4y^2 = 9x^2$, $x = \sqrt{2}$, $y = \frac{3\sqrt{2}}{2}$, so $f = 18$.
4	**(e)**	First octant of ball $\rho \le 3$: $\theta, \phi \in [0, \frac{\pi}{2}]$, integrand $\rho \cdot \rho^2 \sin\phi$.
5	**(d)**	$\nabla F \parallel \mathbf{i} \Rightarrow b = c = 0$; $4a^2 = 100$, so $a^2 + b^2 + c^2 = 25$.
6	**(a)**	Critical points $(0,1)$ ($D = -16$, saddle) and $(\frac{4}{3}, \frac{7}{3})$ ($D = 16$, $f_{xx} = 8$, min).
7	**(b)**	$4\sqrt{2} \cdot \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = 2$.
8	**(e)**	Cardioid $\frac{3\pi}{2}$ minus circle $\frac{\pi}{4} = \frac{5\pi}{4}$.
9	**(a)**	$D = \|\overrightarrow{PQ} \times \mathbf{u}\|/\|\mathbf{u}\| = 3/3 = 1$.
10	**(e)**	Region $0 \le x \le y \le 4$: $\int_0^4 \int_x^4 f\, dy\, dx$.
11	**(e)**	Interior $f(0,1) = -4$; boundary $y = 2$ gives $4x^2 \le 4$; $y = 2x^2$ gives at most 4.
12	**(d)**	Minimize d^2: $x = y = 1$, point $(1,1,2)$, so $1 + 1 + 8 = 10$.
13	**(e)**	$z = (9 - x)/4$ is largest at smallest x; $y^2 \ge 0$ forces $x \ge -3$: $(-3, 0, 3)$.
14	**(a)**	Base disk $(x-1)^2 + y^2 \le 1$, height $2x - x^2 - y^2$ (paraboloid minus plane).
15	**(d)**	Distance $= 4\|t\| = 8$, $t = 2$: point $(7, 1, 10)$, so $49 + 1 + 100 = 150$.
16	**(d)**	$\rho = 4$, $\theta = \frac{\pi}{6}$, $\cos\phi = \frac{\sqrt{3}}{2}$ so $\phi = \frac{\pi}{6}$.
17	**(c)**	z gives 10; swap order: $10 \int_0^{\sqrt{\pi/2}} y \sin y^2\, dy = 5$.
18	**(b)**	Speed $= \sqrt{2}\, ae^{-t}$, so $L = \sqrt{2}\, a(1 - e^{-\pi/2})$; $\sqrt{2}\, a = 4$ gives $a = \sqrt{8}$.
19	**(d)**	$\partial w/\partial t = 2t + 4s^2t^3 = 4 + 288 = 292$.
20	**(d)**	Spherical: $\rho \le 2$, $\phi \le \frac{\pi}{4}$: $V = 2\pi \cdot \frac{8}{3}(1 - \frac{\sqrt{2}}{2}) = \frac{8\pi}{3}(2 - \sqrt{2})$.
21	**(b)**	$\nabla h(500, 300) = \langle -1, -2.4 \rangle$; $-\nabla h \parallel 5\mathbf{i} + 12\mathbf{j}$.